# Alert Reporting

Alert reporting is process of documenting security event in a clear, structured and professional way so that others in the soc can understand

- What happens
- How serious it is
- What evidence was found
- What action was taken
- And what should happen next

A good report help Tier 2, Incident responders, and manger makes quick decisions

**Why it matters**

- Create a record of the alert, Raw SIEM logs stored for 3-12 months, but alerts are kept indefinitely as a result, its better to keep all the context inside the alert just in case
- Supports incident escalation
- Help track analyst decisions
- Improves communication between teams
- Protects the SOC with evidence and audit trail
- Provide context for escalation, A well written report saves a lot of time for L2 analysts, Also it helps them quickly understand what happened
- Improve investigation skills if you can't explain it simply you don't understand it well enough, Report writing is a great way to boost L1 skills by summarizing alerts

**Report Format**

Imagine yourself as an L2 analyst, a DFIR team member, or an IT professional who needs to understand the alert. What would you want to see in the report?

Five W's is the recommended approach and include at least these items in the report

**Who:** Which user logs in, runs the command, or downloads the file

**What:** What exact action or event sequence was performed

**When:** When exactly did the suspicious activity start and ended

**Where:** Which Device, IP, or website was involved in the alert

**Why:** The most important W, the reasoning for you final verdict

**SOC alert should include**

1. **Alert Details**

- Alert Name
- Alert ID
- Date and Time
- Source tool or SIEM
- Affected system or user
- Severity level

2. **Summary**
   A short/Brief/concise summary of what happened
3. **Evidence / Findings**
   - IP Addresses
   - Usernames
   - Hostnames
   - File names
   - URLs
   - Hashes
   - Event IDs
   - Timestamps

4. **Analysis**
   - Benign,
   - Suspicious
   - Malicious
5. **Impact**
   - Account compromise
   - Malware execution
   - Data access
   - Lateral movement
   - Phishing exposure
6. **Actions taken**
   - Checked logs
   - Searched related events
   - Validated against threat intel
   - Reset account
   - Isolated host
   - Escalated to IR
7. **Recommendation/ next step**
   - Monitor the host
   - Block the IP

- Reset password
- Open incident ticket
- Investigate further

**Alert Report Checklist**

- All 5 W's are included in the report
- Alert report is clear and precise
- Alert report forms a single timeline
- Evidence and indicators are attached
- Escalation reason to L2 is explained

## Reporting Guide

| Mar 27th 2025 at 19:25 | Email Marked as Phishing after Delivery | High | Closed | True Positive | E.Fleming (L2) |
|---|---|---|---|---|---|

| | |
|---|---|
| Description: | The email was classified as phishing post-delivery after an automated analysis. If the email is spoofed or contains any suspicious links or files, it must be deeply investigated. |
| Subject: | Important Update: Microsoft Teams Pricing Increase |
| Body Keywords: | 600% price increase; urgent notice; download the report; read the details; |
| Sender: | Microsoft Support<support@microsoft.com> |
| Recipient: | Eddie Huffman, IT Manager<e.huffman@tryhackme.thm> |
| Security Checks: | SPF/Fail; DKIM/Fail; |
| Attached URLs: | None |
| Attached Files: | REPORT.rar |
| Comment: | On 2025-03-27 19:25, Eddie Huffman (IT Manager) received an email from support@microsoft.com with subject referencing an urgent report and attachment REPORT.rar. Automated analysis flagged it as phishing; SPF: Fail and DKIM: Fail. Message quarantined; attachment exported to isolated sandbox for analysis. Please escalate to L2 for IOC extraction, endpoint forensics if opened, and block sender/domain at gateway. |

Edit Alert

Email Marked as Phishing after Delivery

Status: In Progress

Verdict: True Positive

Severity: High

Assignee: E.Fleming (L2)

Analyst Comment *

On 2025-03-27 19:25, Eddie Huffman (IT Manager) received an email from support@microsoft.com with subject referencing an urgent report and attachment REPORT.rar. Automated analysis flagged it as phishing; SPF: Fail and DKIM: Fail. Message quarantined; attachment exported to isolated sandbox for analysis. Please escalate to L2 for IOC extraction, endpoint forensics if opened, and block sender/domain at gateway.

Authenticity: 100%

Characters: 413/50

Save

On 2025-03-27 19:25, Eddie Huffman (IT Manager) received an email from support@microsoft.com with

subject referencing an urgent report and attachment REPORT.rar. Automated analysis flagged it as phishing; SPF: Fail and DKIM: Fail. Message quarantined; attachment exported to isolated sandbox for analysis. Please escalate to L2 for IOC extraction, endpoint forensics if opened, and block sender/domain at gateway.

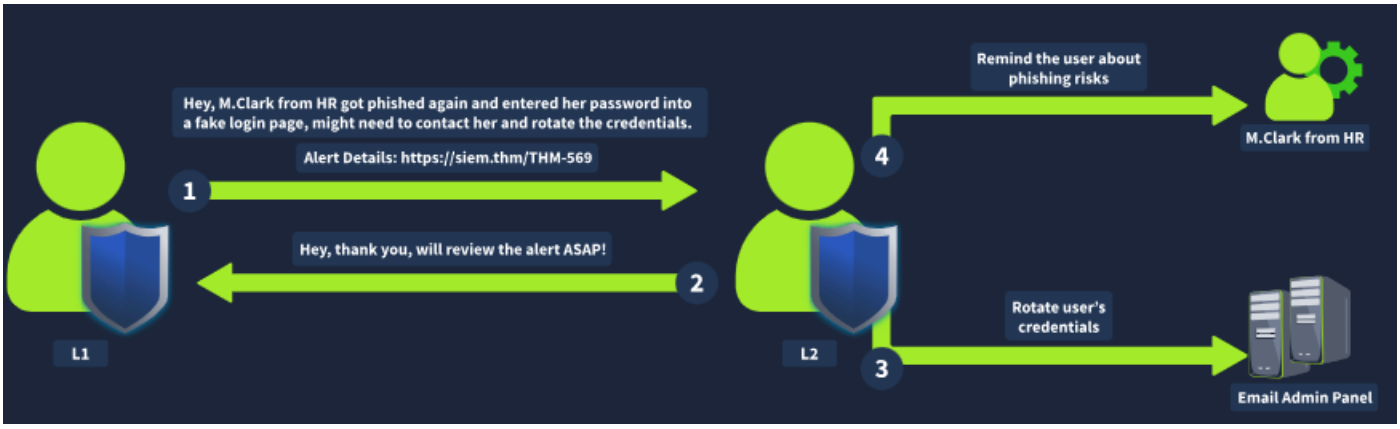
## Escalation Guide

After you have made a verdict and written your alert report, you must choose whether to escalate the alert to L2. Again, the answer may differ from team to team, but the following recommendations would generally fit most SOC teams. You should escalate the alerts if:

1. The alert is an indicator of a major cyberattack requiring deeper investigation or DFIR
2. Remediation actions like malware removal, host isolation, or password reset are required
3. Communication with customers, partners, management, or law enforcement agencies is required
4. You just do not fully understand the alert and need some help from more senior analysts

Escalation Steps

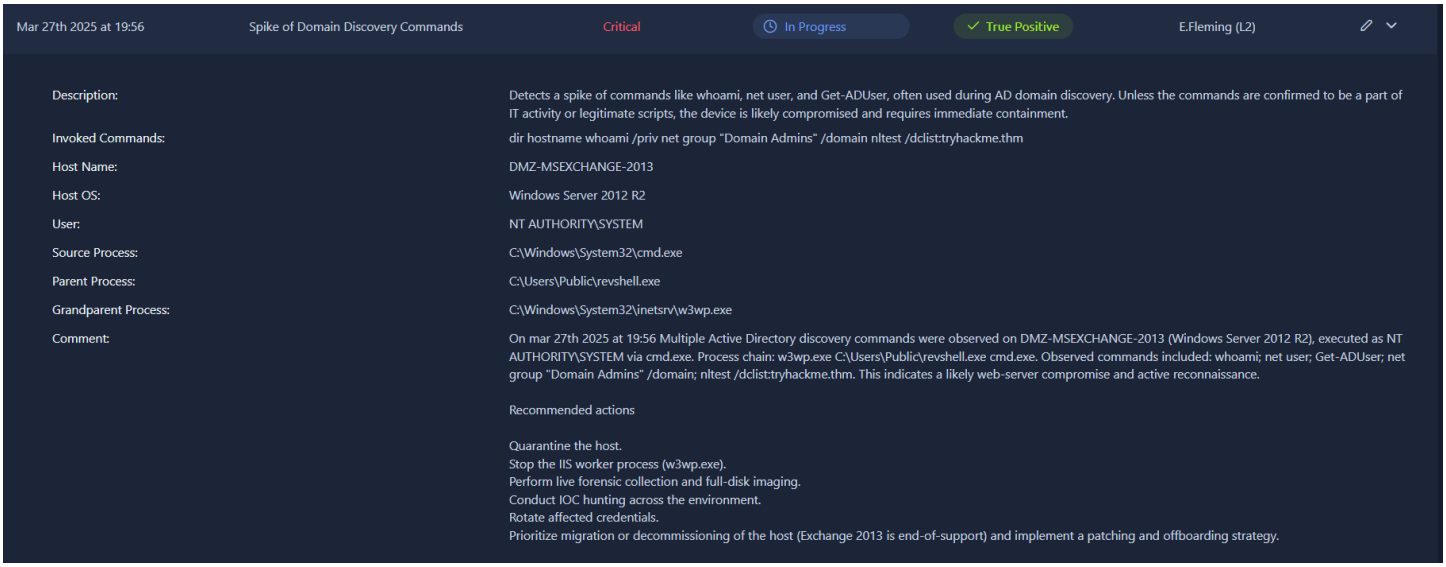
To escalate the alert, in most cases, all you have to do is to **reassign the alert to the L2 on shift** and ping them in corporate chat or in person. In some teams though, you may be required to create a formal written escalation request with dozens of required fields.




No matter what the agreements are, L2 will eventually receive the ticket from you, read your report, and contact you in case of any questions. Once everything is clear, the L2 analyst will typically research the alert details further, validate if the alert is indeed a True Positive, communicate with other departments if needed, and, for major incidents, start a formal Incident Response process.

Requesting L2 Support

It is generally fine for L1 to request senior support if something is unclear. Especially in your first months, it's always better to discuss the alert and clarify SOC procedures than to blindly close the alert you don't understand yourself. The procedures for requesting support may differ, but the flow generally looks like this:




On mar 27th 2025 at 19:56 Multiple Active Directory discovery commands were observed on DMZ-MSEXCHANGE-2013 (Windows Server 2012 R2), executed as NT AUTHORITY\SYSTEM via cmd.exe. Process chain: w3wp.exe C:\Users\Public\revshell.exe cmd.exe. Observed commands included: whoami; net user; Get-ADUser; net group "Domain Admins" /domain; nltest /dclist:tryhackme.thm. This indicates a likely web-server compromise and active reconnaissance. Recommended actions Quarantine the host. Stop the IIS worker process (w3wp.exe). Perform live forensic collection and full-disk imaging. Conduct IOC hunting across the environment. Rotate affected credentials. Prioritize migration or decommissioning of the host (Exchange 2013 is end-of-support) and implement a patching and offboarding strategy.

**Note:** Some processes may be unfamiliar to an SOC analyst. Not recognizing a process does not mean you are a bad analyst. there are thousands of processes and you cannot be expected to know them all. When you encounter an unfamiliar process, search for it; that search is part of your investigation.

## SOC Communication

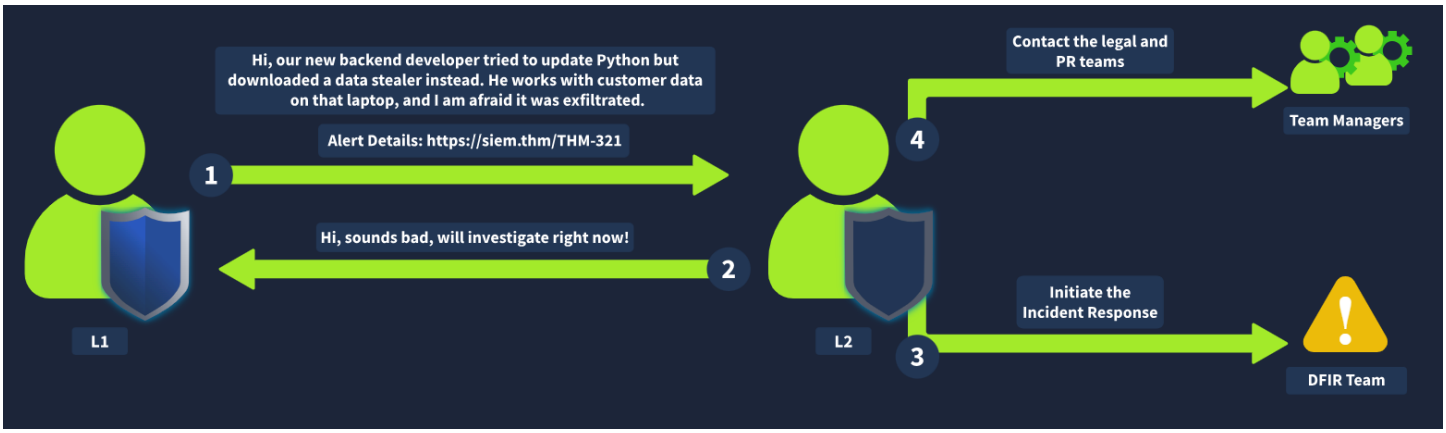
The escalation and reporting topics should sound straightforward and logical to you. But, as always, it's easier said than done, and you should be prepared for unexpected scenarios and know what to do in

critical cases. In the best scenario, the SOC team has its own **Crisis Communication** procedures - the guides and processes to help you and your teammates resolve the issues. If not, you are advised to read the cases below and be prepared to handle them effectively.

Communication Cases

- **You need to escalate an urgent, critical alert, but L2 is unavailable and does not respond for 30 minutes.**
  Ensure you know where to find emergency contacts. First, try to call L2, then L3, and finally your manager.
- **The alert about Slack/Teams account compromise requires you to validate the login with the affected user.**
  Do not contact the user through the breached chat - use alternative contact methods like a phone call.
- **You receive an overwhelming number of alerts during a short period of time, some of which are critical.**
  Prioritise the alerts according to the workflow, but inform your L2 on shift about the situation.
- **After a few days, you realise that you misclassified the alert and likely missed a malicious action.**
  Immediately reach out to your L2 explaining your concerns. Threat actors can be silent for weeks before impact.
- **You can not complete the alert triage since the SIEM logs are not parsed correctly or are not searchable.**
  Do not skip the alert - investigate what you can and report the issue to your L2 on shift or SOC engineer.

Communication By L2




**Images are taken From Tryhackme**